HW3

Dongwook Hwang

Q1.

r \ c	Running Man (러닝맨; RM)		I Live Alone (나 혼자 산다)		Hangingout with Yoo (놀면뭐하니; HWY)		I am Solo (나는 솔로; IAS)		marginal	student ratio
Freshman	0.0225	0.15	0.0375	0.25	0.045	0.3	0.045	0.3	0.15	0.15
Sophomore	0.02	0.1	0.06	0.3	0.04	0.2	0.08	0.4	0.2	0.2
Junior	0.025	0.1	0.05	0.2	0.075	0.3	0.1	0.4	0.25	0.25
Senior	0.24	0.6	0.04	0.1	0.08	0.2	0.04	0.1	0.4	0.4
marginal P(c) =	0.3075		0.1875		0.24		0.265		1.0	

P(c) = marginal (TV show)

P(r) = marginal (college year)

If the college year and the favorite TV show are independent, p(r, c) = p(r) x P(c)

Therefore, the college year and the favorite TV show is not independent.

Q2.

actual ⟶ kit result ⊕ → ⊕ : 90%

hit rate : 90% → positive accuracy : 90%
a person has COVID-19, the test result is positive 90%
false alarm (FA) rate : 4% ⊖ ⟶ ⊕ : 4%

Q2-1. base rate of the disease : 0.0004
sample : random in Korea.
↳ kit result : positive

test result	disease: present (Ⓟ)	disease: absent (ⓐ)	marginal (test result) P(r)	
⊕	0.0004 × 0.9	(1-0.0004) × 0.04	0.040344	P(⊕)
⊖	0.0004 × (1-0.9)	(1-0.0004) × (1-0.04)	0.959656	P(⊖)
marginal (disease) P(c)	0.0004 P(Ⓟ)	0.9996 P(ⓐ)	1.0	

∴ posterior probability = P(Ⓟ|⊕) / P(⊕) = 0.0004 · 0.9 / 0.040344
prior = 0.0004 ≈ 0.89%

Q2-2. Ⓟ ⟶ ⊖ : 10%
P(⊖|Ⓟ) = 0.1 ⟶ hit rate : 90%
P(⊕|Ⓟ) = 0.9 so, it is totally same with Q2-1
∴ 0.89%

Q2-3. base rate : 0.05
false negative : 0.1, FA rate : 0.04 → All of Q2 kits are same.

P(⊕) = 0.05 · 0.9 + 0.95 · 0.04 = 0.083
posterior probability : P(Ⓟ|⊕) / P(⊕) = 0.05 · 0.9 / 0.083 ≈ 54%